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Preprocessing for image classification by convolutional neural networks

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Abstract:

In recent times, the Convolutional Neural Networks have become the most powerful method for image classification. Various researchers have shown the importance of network architecture in achieving better performances by making changes in different layers of the network. Some have shown the importance of the neuron's activation by using various types of activation functions. But here we have shown the importance of preprocessing techniques for image classification using the CIFAR10 dataset and three variations of the Convolutional Neural Network. The results that we have achieved, clearly shows that the Zero Component Analysis(ZCA) outperforms both the Mean Normalization and Standardization techniques for all the three networks and thus it is the most important preprocessing technique for image classification with Convolutional Neural Networks.

Published in: 2016 IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)

Date of Conference: 20-21 May 2016

DOI: 10.1109/RTEICT.2016.7808140

Date Added to IEEE Xplore: 09 January 2017

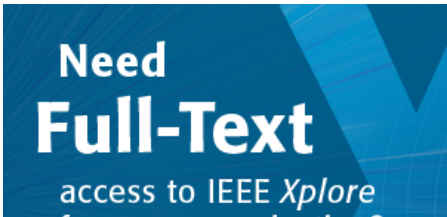
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Conference Location: Bangalore, India

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